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1 C:\Users\270708326\PycharmProjects\PythonProject\SmartHealthCare\.venv\Scripts\python.exe C:\
  Users\270708326\PycharmProjects\PythonProject\SmartHealthCare\predictive_modeling.py
2 =====
3 SMARTCARE PREDICTIVE MODELING & CLUSTERING
4 =====
5
6 ✓ Data loaded: 50 records
7
8 =====
9 1. WEEKLY HEALTH SCORE PREDICTION MODEL
10 =====
11
12 Training set: 35 samples
13 Testing set: 15 samples
14
15 --- Multiple Linear Regression ---
16 RMSE: 17.317
17 MAE: 14.131
18 R2 Score: -0.735
19
20 Feature Coefficients:
21   Gender_Male: -4.594
22   SleepDuration_Hours: 1.014
23   Age: 0.205
24   SleepQualityScore: 0.196
25   HeartRate_Avg: 0.185
26   StressIndex: -0.044
27   CalorieBurn: 0.007
28   DailyStepCount: -0.000
29
30 --- Random Forest Regression ---
31 RMSE: 15.192
32 MAE: 12.094
33 R2 Score: -0.335
34
35 Feature Importance:
36   CalorieBurn: 0.256
37   SleepQualityScore: 0.252
38   SleepDuration_Hours: 0.140
39   DailyStepCount: 0.117
40   HeartRate_Avg: 0.082
41   StressIndex: 0.080
42   Age: 0.057
43   Gender_Male: 0.015
44
45 Cross-Validation R2 Scores: [-0.60669118 -0.54493798 -0.22311134 -0.20770697 -0.43540837]
46 Mean CV R2: -0.404 (+/- 0.326)
47
48 ✓ Health score prediction visualization saved
49
50 =====
51 2. FALL RISK PREDICTION MODEL
52 =====
53
54 Class Distribution:
55   No Falls: 45 (90.0%)
56   Falls: 5 (10.0%)
57
58 --- Logistic Regression ---
59 Accuracy: 0.867
60 AUC-ROC: 0.357
61
62 Classification Report:
63           precision    recall  f1-score   support
64
65    No Fall         0.93      0.93      0.93        14
66      Fall         0.00      0.00      0.00         1
67
68    accuracy                   0.87        15
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69     macro avg           0.46      0.46      0.46      15
70 weighted avg           0.87      0.87      0.87      15
71
72
73 Feature Coefficients (Fall Risk Factors):
74 SleepDuration_Hours: 0.534 (↑ increases fall risk)
75 Age: -0.169 (↓ decreases fall risk)
76 SleepQualityScore: -0.037 (↓ decreases fall risk)
77 StressIndex: 0.018 (↑ increases fall risk)
78 WeeklyHealthScore: -0.005 (↓ decreases fall risk)
79 DailyStepCount: 0.001 (↑ increases fall risk)
80
81 --- Random Forest Classifier ---
82 Accuracy: 0.933
83 AUC-ROC: 0.714
84
85 Classification Report:
86               precision    recall  f1-score   support
87
88    No Fall         0.93      1.00      0.97      14
89     Fall         0.00      0.00      0.00       1
89
90
91    accuracy                0.93      15
92    macro avg         0.47      0.50      0.48      15
93 weighted avg         0.87      0.93      0.90      15
94
95
96 ✓ Fall risk prediction visualization saved
97
98 =====
99 3. K-MEANS CLUSTERING - PATIENT SEGMENTATION
100 =====
101
102 --- Elbow Method for Optimal K ---
103 K=2: Inertia=201.46, Silhouette Score=0.170
104 K=3: Inertia=170.33, Silhouette Score=0.176
105 K=4: Inertia=139.90, Silhouette Score=0.205
106 K=5: Inertia=126.34, Silhouette Score=0.207
107 K=6: Inertia=109.74, Silhouette Score=0.205
108 K=7: Inertia=99.06, Silhouette Score=0.215
109
110 ✓ Selected K = 4 clusters
111
112 Cluster Distribution:
113 Cluster 0: 13 patients (26.0%)
114 Cluster 1: 12 patients (24.0%)
115 Cluster 2: 10 patients (20.0%)
116 Cluster 3: 15 patients (30.0%)
117
118 --- Cluster Profiles ---
119      Age  DailyStepCount  ...  Risk_Score  FallAlerts
120 Cluster
121 0      56.85      7352.15  ...      58.92      0.15
122 1      69.08      4836.33  ...      53.49      0.08
123 2      67.40      9915.70  ...      26.35      0.10
124 3      74.40      8089.47  ...      63.32      0.07
125
126 [4 rows x 7 columns]
127
128 Cluster Interpretation:
129
130 Cluster 0:
131 Average Age: 56.8 years
132 Activity: 7352 steps/day
133 Sleep: 5.6 hours/day
134 Stress: 66.6/100
135 Health Score: 79.8/100
136 Risk Score: 58.9/100
137 Fall Rate: 15.38%
```

138 Label: Moderate Risk - General Monitoring

139

140 Cluster 1:

141 Average Age: 69.1 years

142 Activity: 4836 steps/day

143 Sleep: 8.0 hours/day

144 Stress: 56.8/100

145 Health Score: 88.4/100

146 Risk Score: 53.5/100

147 Fall Rate: 8.33%

148 Label: Moderate Risk - General Monitoring

149

150 Cluster 2:

151 Average Age: 67.4 years

152 Activity: 9916 steps/day

153 Sleep: 8.0 hours/day

154 Stress: 36.5/100

155 Health Score: 75.2/100

156 Risk Score: 26.3/100

157 Fall Rate: 10.00%

158 Label: Moderate Risk - General Monitoring

159

160 Cluster 3:

161 Average Age: 74.4 years

162 Activity: 8089 steps/day

163 Sleep: 6.4 hours/day

164 Stress: 71.8/100

165 Health Score: 69.5/100

166 Risk Score: 63.3/100

167 Fall Rate: 6.67%

168 Label: High Risk - Needs Intervention

169

170 ✓ Clustering visualization saved

171

172 =====

173 4. RISK SCORE VALIDATION

174 =====

175

176 Health Outcomes by Risk Category:

177

178 Low Risk (n=6):

179 Avg Health Score: 87.0

180 Avg Age: 70.0

181 Fall Rate: 0.0%

182 High Stress (>70): 0 (0.0%)

183 Poor Sleep (<6h): 0 (0.0%)

184

185 Medium Risk (n=18):

186 Avg Health Score: 81.3

187 Avg Age: 66.0

188 Fall Rate: 11.1%

189 High Stress (>70): 4 (22.2%)

190 Poor Sleep (<6h): 6 (33.3%)

191

192 High Risk (n=19):

193 Avg Health Score: 75.3

194 Avg Age: 67.3

195 Fall Rate: 15.8%

196 High Stress (>70): 10 (52.6%)

197 Poor Sleep (<6h): 4 (21.1%)

198

199 Critical Risk (n=7):

200 Avg Health Score: 68.4

201 Avg Age: 67.3

202 Fall Rate: 0.0%

203 High Stress (>70): 6 (85.7%)

204 Poor Sleep (<6h): 4 (57.1%)

205

206 Risk Score Correlations:

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207 Risk Score vs Health Score: r = -0.463
208 Risk Score vs Stress Index: r = 0.569
209 Risk Score vs Age: r = -0.065
210
211 =====
212 5. SAVE ENHANCED DATASET
213 =====
214
215 ✓ Enhanced dataset saved: 'smartcare_with_predictions.csv'
216 Added columns: Cluster, Cluster_Label
217
218 =====
219 MODEL PERFORMANCE SUMMARY
220 =====
221
222           Model           Primary Metric Performance
223 Linear Regression (Health Score)      R² = -0.735      Moderate
224 Random Forest (Health Score)         R² = -0.335          Good
225 Logistic Regression (Fall Risk)      Accuracy = 0.867      Good
226 Random Forest (Fall Risk)           Accuracy = 0.933      Good
227 K-Means Clustering Silhouette = 0.205      Moderate
228
229 =====
230 KEY PREDICTIVE INSIGHTS
231 =====
232
233 1. HEALTH SCORE PREDICTION:
234   • Best model: Random Forest (R² = -0.335)
235   • Top 3 predictors: CalorieBurn, SleepQualityScore, SleepDuration_Hours
236   • Model can explain -33.5% of health score variation
237
238 2. FALL RISK PREDICTION:
239   • Model accuracy: 93.3%
240   • Key risk factors: Age, WeeklyHealthScore, DailyStepCount
241   • AUC-ROC: 0.714
242
243 3. PATIENT SEGMENTATION:
244   • Optimal clusters: 4
245   • Silhouette score: 0.205
246   • Clusters enable targeted intervention strategies
247
248 =====
249 PREDICTIVE MODELING COMPLETE
250 =====
251
252 Generated Outputs:
253   1. health_score_prediction.png
254   2. fall_risk_prediction.png
255   3. clustering_analysis.png
256   4. smartcare_with_predictions.csv
257
258 Next Steps:
259   • Create Power BI dashboard
260   • Develop intervention recommendations
261   • Document methodology in report
262 =====
263
264 Process finished with exit code 0
265
```